

SLSeeds System Report

1. System Overview

SLSeeds is a fully functional, modern e-commerce platform designed to sell seeds and agricultural products online. The system is built with a strong focus on real-time communication, secure payments, and easy management for both the business owner and the customers.

2. Technology Stack

Backend (Server-Side)

- **Framework:** Laravel (PHP Framework)
- **Language:** PHP 8.x
- **Database:** MySQL
- **Real-time Communication:** Laravel Reverb (WebSockets) for instant messaging without page reloads.

Frontend (Client-Side)

- **Design & Styling:** Tailwind CSS (for modern, responsive, and mobile-friendly designs)
- **Interactivity:** Alpine.js (Used for the live chat UI and dropdowns) & JavaScript
- **Templates:** Laravel Blade
- **Popups & Alerts:** SweetAlert2 (for beautiful confirmation boxes)

Key Integrations & Tools

- **Payment Gateway:** PayHere API (Supports Credit/Debit cards and local payment methods)

- **Document Generation:** DOMPDF (Used to generate downloadable Order Receipts and Admin Reports)
- **Email System:** Standard SMTP Integration (For User Registration Verification and Notifications)

3. Customer Features (User Panel)

Authentication & Security

- **Strict Registration:** Users must register with a valid email address.
- **Email Verification:** A secure verification link is sent to the user's email. They cannot access the dashboard or place orders until they click the link.
- **Profile Management:** Users can update their name, delivery address, and phone number (with strict 10-digit format validation).

Shopping Experience

- **Multi-Language Support:** Customers can view product names and descriptions in both English and Sinhala.
- **Shopping Cart:** Users can add items, update quantities, and remove items. The system automatically prevents adding more items than the available stock.
- **Checkout System:** Option 1: Cash on Delivery (COD). Option 2: Online Payment via PayHere.

Order & Support Management

- **Order Tracking:** Customers can view their past orders, check the current status, and download PDF Order Receipts.
- **Live Chat Widget:** A floating green chat bubble allows customers to talk to the Admin in real-time and even send images.

4. Admin Features (Admin Panel)

Dashboard & Analytics

- **Overview:** Quick view of total sales, total orders, active customers, and low stock items.
- **Notifications:** Real-time bell notifications alert the admin when a new order is placed or when a product's stock runs low.

Catalog & Stock Management

- **Product Management:** Admin can add new products, upload images, set prices, and add English/Sinhala translations.
- **Stock Tracking:** Stock quantities decrease automatically when an order is placed. Admin can also manually adjust stock levels.

Order Management & Reporting

- **Order Processing:** Admin can view full order details and update the status.
- **Manual Orders:** Admin can manually place orders on behalf of customers who call over the phone.
- **Date Filtering & Exports:** Admin can filter orders by a specific date and download the data as a CSV (Excel) or PDF file.

Customer & Chat Management

- **Customer List:** Admin can view all registered users and manually update their details.
- **Real-Time Admin Chat:** A dedicated Chat Page allows the admin to reply to customers instantly with Optimistic UI, Image Attachments, and Deletion features.

Financial Management

- **Financial Reports:** Calculates total revenue, separates COD vs. Online payments, handles refunds, and exports reports to PDF or Excel.